



NVDEC Application Note

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Chapter 1. Introduction

NVIDIA GPUs and NVIDIA Jetson platforms contain a hardware-based decoder (referred to as NVDEC in this document) which provides fully accelerated hardware-based video decoding for several popular codecs. With complete decoding offloaded to NVDEC, the graphics engine and CPU are free for other operations.

NVDEC supports much faster than real-time decoding which makes it suitable for transcoding scenarios in addition to video playback.

The hardware capabilities available in NVDEC are exposed through APIs referred to as NVDECODE APIs in this document. This document provides information about the capabilities of the NVDEC engine and the features exposed through NVDECODE APIs.

Chapter 2. NVDEC Capabilities

At a high level, *NVDEC Hardware Capabilities* summarizes the capabilities of the NVDEC engine exposed through NVDECOD API.

Table 1: NVDEC Hardware Capabilities

Hardware Features	Tur- ing/GA100/Hopp GPUs	GA10x ³ Ada GPUs	and Blackwell GPUs/ son Thor	Jet-
VC1 Simple, Main & Advanced profiles	Y	Y	Y	
MPEG4 Simple and Advanced Simple Pro- files	Y	Y	Y	
MPEG2 Simple & Main profiles	Y	Y	Y	
H.264 Baseline, Main, High Profiles	Y	Y	Y	
H.264 8192x8192 Decoding support	N	N	Y	
H264 High10/High422 profiles	N	N	Y	
VP8	Y	Y	Y	
HEVC Main and Main 10 Profile ¹	Y	Y	Y	
HEVC 444 decoding	Y	Y	Y	
HEVC main 422 10/12 profiles	N	N	Y	

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Hardware Features	Tur- ing/GA100/Hopp GPUs	GA10x ³ Ada GPUs	and Blackwell GPUs/ son Thor	Jet-
HEVC 8192x8192 Decoding support	Y	Y	Y	
VP9 8192x8192 Decoding support	Y	Y	Y	
VP9 Profile 0 ¹	Y	Y	Y	
Multiple NVDECs ²	Y	Y	Y	
AV1 Main Profile decoding	N	Y	Y	

► **Y:** Supported, **N:** Unsupported

► ¹: Present in select GPUs

► ²: Present in select GPUs

► ³: GA10x GPUs include all GPUs based on Ampere architecture except GA100

Chapter 3. NVDEC Performance

NVDEC natively supports multiple hardware decoding contexts with negligible context-switching penalty. As a result, subject to the hardware performance limit and available memory, an application can decode multiple videos simultaneously.

The hardware and software maintain the context for each decoding session, allowing many simultaneous decoding sessions to run in parallel with minimal context switch penalty. *NVDEC Decoding Performance* provides indicative data of the decoding performance of NVDEC in GPUs based on Turing, Ampere, Ada, and Blackwell architectures as well as Jetson platforms based on Thor for AV1, HEVC, VP9, and H.264 encoded bitstreams. The performance varies across GPU classes (e.g. Quadro, Tesla), and scales (almost) linearly with the clock speeds for each hardware.

Table 1: NVDEC Decoding Performance (Indicative)

GPU Architecture/ Jetson Platform	Codec	Performance in frames/second
Turing	H.264	771
	VP9	932
	VP9 10 bit	925
	HEVC	1316
	HEVC Main10	1158
Ampere	H.264	748
	VP9	1075
	VP9 10 bit	1120
	HEVC	1415
	HEVC Main10	1299
Ada	AV1	790
	H.264	903
	VP9	1290
	VP9 10 bit	1342
	HEVC	1641
Blackwell	HEVC Main10	1520
	AV1	1018
	H.264	2172
	VP9	1445
	VP9 10 bit	1498
Jetson Thor	HEVC	1872
	HEVC Main10	1818
	AV1	1119
	H.264	1434
	VP9	1019
	VP9 10 bit	1016
	HEVC	1293
	HEVC Main10	1130
	AV1	794

► All the measurement is done on the highest video clocks as reported by nvidia-smi (i.e. 1860

MHz, 1665 MHz, 2160 MHz, 2362 MHz for Turing, Ampere, Ada, and Blackwell GPUs respectively and 1691 MHz for Jetson Thor platform). The performance should scale according to the video clocks as reported by nvidia-smi on target GPU. Information on nvidia-smi can be found at <https://developer.nvidia.com/nvidia-system-management-interface>.

- ▶ Resolution/Input format: 1920x1080/YUV 4:2:0
- ▶ Software: Windows 11/Jetson Linux 38.2 for GPUs/Jetson platform, Video Codec SDK v13.1
- ▶ Hopper and GA100 GPUs contain NVDEC with same architecture as Turing. As a result, the decoding performance on Hopper and GA100 GPUs is the same as that of Turing GPUs, scaled by the clock speed. To view the clocks available on your GPU, please use the tool nvidia-smi included with the NVIDIA driver.

Some NVIDIA GPUs may have multiple NVDEC engines per chip. Refer to the NVIDIA Video Codec SDK Support Matrix for details. This increases the aggregate decoder performance of the GPU. NVIDIA driver takes care of load balancing among multiple NVDEC engines on the chip, so that applications don't require any special code to take advantage of multiple decoders and automatically benefit from higher decoder capacity on higher-end GPU hardware. The decode performance listed in *NVDEC Decoding Performance* is given per NVDEC engine. Thus, if a GPU has multiple NVDECs, multiply the corresponding number in *NVDEC Decoding Performance* by the number of NVDECs per chip to get aggregate maximum performance (applicable only when running multiple simultaneous decode sessions). Note that performance with a single decoding session cannot exceed performance per NVDEC, regardless of the number of NVDECs present on the GPU.

Chapter 4. Programming NVDEC

The NVDEC API provides access to the video decoding features of NVDEC described in the previous chapters. The API includes implementations of commonly used post-processing operations such as scaling, cropping, aspect ratio conversion, deinterlacing and color space conversion to the decoded output.

Refer to the SDK release notes for information regarding the required driver version.

Various capabilities of NVDEC are exposed to the application software via the NVIDIA proprietary application programming interface (NVDEC APIs). Refer to the Video Decoder Programming guide for details on using these APIs.

For a complete list of GPUs supporting hardware accelerated decoding refer to <https://developer.nvidia.com/nvidia-video-codec-sdk>.

Chapter 5. FFmpeg Support

FFmpeg is the most popular multimedia transcoding tool used extensively for video and audio transcoding. Note that FFmpeg is an open-source project and its usage is governed by specific licenses and terms and conditions.

The video hardware accelerators in NVIDIA GPUs can be effectively used with FFmpeg to significantly speed up the video decoding, encoding and end-to-end transcoding at very high performance.

Note

FFmpeg with HW acceleration is supported on the Jetson Thor platform (including Jetson T4000 and T5000 modules) with JetPack 7.1 and later.

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